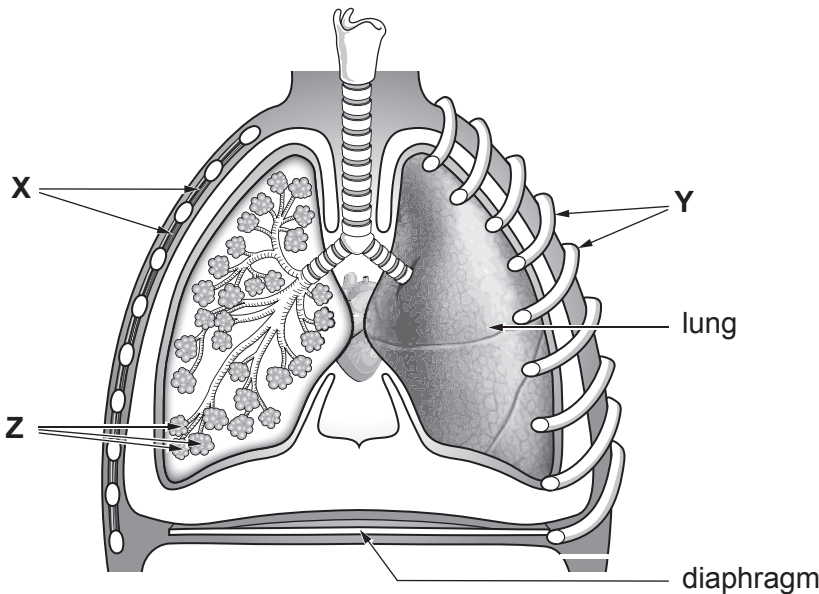


Answer all questions.

1. (a) The diagram shows the human thorax after breathing in.



- (i) State the name of the structures labelled:
- X [1]
- Y [1]
- (ii) The structures labelled Z are the sites of gas exchange.
State the name of the structures labelled Z. [1]
-
- (iii) Complete the following sentences by choosing the correct words from the list. [2]

decreases contracts relaxes increases

When we breathe **out**, the diaphragm and becomes
dome shaped. As a result, pressure in the thorax , so
air is forced out of the lungs.

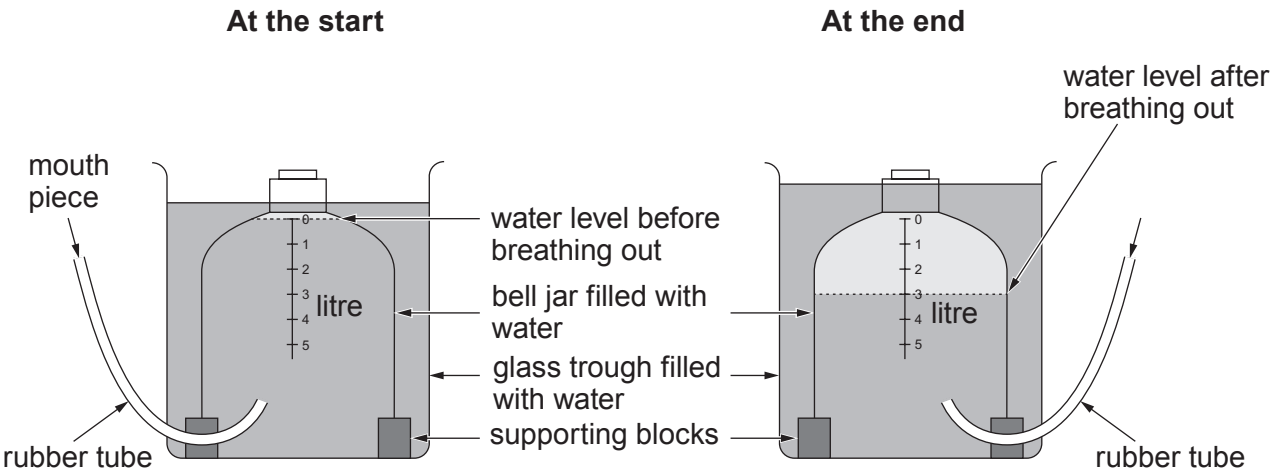


(iv) Choose the letter (**A-C**) that shows the pathway of air from the lungs when we breathe **out**. [1]

- A** trachea → bronchus → bronchiole
- B** bronchus → trachea → bronchiole
- C** bronchiole → bronchus → trachea

Answer letter =

(b) The maximum volume of air that can be breathed out in one breath is called the vital capacity. The diagram shows apparatus that Lowri used to measure her vital capacity.



Describe how Lowri used the apparatus above to measure her vital capacity. [2]

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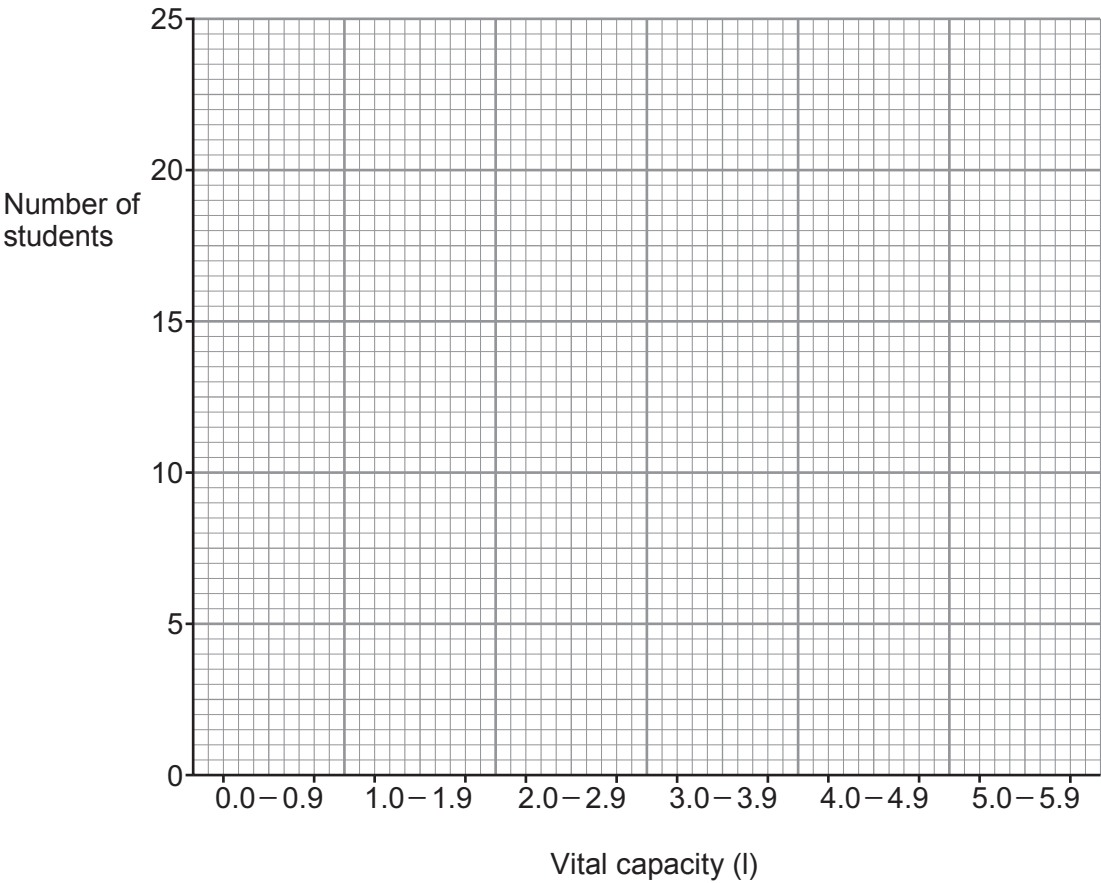
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- (c) The table shows the vital capacities of some students in a school. The vital capacities were measured to the nearest 0.1 litre.

Vital capacity (l)	Number of students
0 - 0.9	5
1.0 - 1.9	9
2.0 - 2.9	20
3.0 - 3.9	8
4.0 - 4.9	23
5.0 - 5.9	7

- (i) Use a ruler to draw a bar chart of the results on the grid. [2]



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(ii) Suggest **two** possible reasons to account for the variation in vital capacity. [2]

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